



2DP-FHS: 2D Pareto Optimized Fog Head Selection for Multiple EEG Healthcare Data Analysis and Computations

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Abstract. In recent years, the increase in healthcare data demands the adoption of fog computing for fast processing and analysis of large data volumes. Fog computing enables real-time data computations that serves closely to healthcare devices. This study proposes the analysis and computations of EEG healthcare data using Pareto optimization-based fog computing. The proposed 2DP-FHS technique selects a fog head among heterogeneous fog devices to manage the EEG data within the fog layer. Additionally, an alternative fog head is designated to ensure continuous operation within the fog layer. The study considers various scenarios and unbiased fog devices in the simulation. Further, it presents the delay analysis varying the number of EEG and fog devices. The results show the effectiveness of the proposed technique across all scenarios highlighting its real-time applicability for various healthcare centers requiring time-critical EEG applications.

Keywords: Pareto optimization · fog head selection · EEG healthcare · delay analysis · fog computing

1 Introduction

The rapid growth in healthcare data necessitates the need for technological advancements in real-time computation and analysis. Healthcare data that provides crucial insights into human physiological conditions are time-sensitive. Electroencephalogram (EEG) signal is one of the important healthcare data that gives vast medical information about human brain conditions [1]. Moreover, the EEG sensors generate continuous signals and a large volume of sensitive data. Analyzing EEG healthcare data is vital for clinical decisions and epidemiological research. Therefore, the adoption of fog computing is vital for efficient EEG data computation, faster decision-making, and minimizing latency. Fog computing is an extension of cloud computing that provides lightweight processing, real-time data analysis, and a limited storage capacity [2, 3]. The fog devices are also capable of handling EEG data generated from multi-channel EEG devices [4].

Moreover, the fog computing architecture is useful for minimizing the delay, energy, and cost using various optimization techniques. Paper [5] proposes a deep learning-based

optimization technique for resource scheduling management in fog computing focusing on minimizing delay and bandwidth. Paper [6] uses FireFly optimization for cost-reduction-based resource allocation and load balancing in cloud fog architecture. Further, multiple heterogeneous fog devices participate in parallel and distributed decision-making processes. Therefore, the optimal selection of fog heads and fog resources is necessary [7–9]. Many techniques are developed by the researchers emphasizing the optimal selection methods for various applications, but there is a lack of developments in EEG healthcare systems. In this work, the authors have focused on the optimal fog head selection technique for EEG based healthcare systems using the Pareto Optimization [10].

1.1 Motivation and Contribution

The existing EEG healthcare systems are mostly cloud-dependent [11]. The cloud provides major services such as big data management, permanent storage, and heavy processing; but consumes high bandwidth and high latency which is unacceptable in real-time healthcare analysis [12, 13]. Moreover, EEG devices are single-user assistive devices and, therefore need dedicated computing resources. To address the above challenges, fog computing is integrated with multiple EEG healthcare systems. The integration of fog computing supports real-time data analysis with minimum latency and faster response time [14, 15]. Further, it supports scalability and heterogeneity which helps in processing huge volumes of EEG data in a distributed manner [16]. The contributions in this paper are listed below:

- i) A two-layer architecture is proposed, where multiple EEG devices are merged in one platform to perform distributed computation and data analytics connecting with a group of heterogeneous fog devices.
- ii) An optimal Fog Head (FH) and an Alternate Fog Head (AFH) are selected using two-dimensional Pareto optimization undertaking multiple scenarios.
- iii) Further delay analysis is performed in the fog layer with a change in the number of EEG devices and the number of fog devices.

The further sections are organized as follows: Sect. 2 describes the proposed methodology. Section 3 illustrates the simulation result. Finally, Sect. 4 concludes the paper.

2 Proposed Methodology

In this paper, the authors have proposed a multi-layer architecture where multiple EEG devices collaborate with multiple fog devices through a fog head. In the real scenario, a single fog device is not sufficient for processing EEG data from multiple EEG devices due to energy, storage, and computational constraints. Therefore, multiple fog devices having different configurations are considered for this work. Also, in this work, an optimal fog device among all the fog devices is selected which is discussed further. The next subsection presents the proposed architectural view and its functions.

2.1 Architectural View

The proposed architecture consists of two layers: a device layer and a fog layer; as shown in Fig. 1. The device layer consists of multiple EEG devices which are independent of each other. In the fog layer, heterogeneous fog devices are connected to the FH.

The functions of the device layer are to:

- Sense brain signals of patients using multiple EEG sensors
- Divide the continuous raw EEG signal into chunks
- Transmit the chunks as output files to the fog layer

The functions of the fog layer are to:

- Receive the EEG data files from the device layer
- Select one of the fog devices as FH
- Collaborative and distributed processing of the tasks among fog devices
- Acknowledge the responses

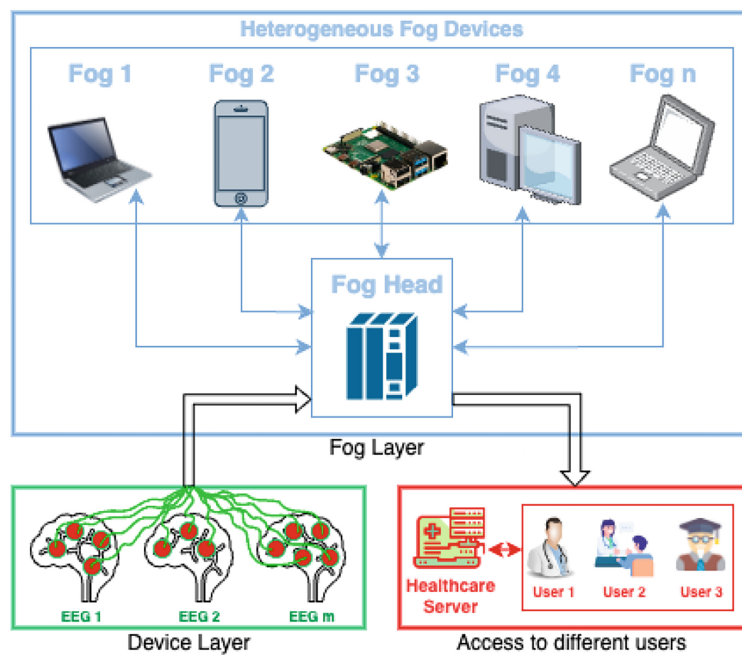


Fig. 1. Proposed 2DP-FHS architecture.

2.2 Fog Head Selection

The necessity of selecting a FH is to collect the EEG files from the device layer and distribute the tasks among fog devices. Also, the additional responsibility of the FH is to

collect the responses generated by the fog devices and to provide access to the end users. This section describes the mathematical modeling of fog parameters and the workflow of the proposed technique.

2.2.1 Mathematical Modeling

In this work, the selection of FH is done by using two parameter index values: (i) Fog Delay Index (FDI) and (ii) Fog Performance Index (FPI).

Theorem 1: The Fog Delay Index (FDI_k) of k^{th} fog device is the ratio between the sum of Propagation Delay (PD_k), Processing Delay (PrD_k), and Average Queuing Delay (AQD_k) to the Total Network Delay (D^{Net}).

Proof:

Definition 1: The propagation delay is the ratio between the summation of $dist_{K,E}^{avg}$, $dist_{K,F}^{avg}$ and the velocity of signal (SS) in wireless medium.

For a set of M number of EEG devices and N number of fog devices, the propagation delay of k^{th} fog device is calculated as:

$$PD_k = \frac{dist_{K,E}^{avg} + dist_{K,F}^{avg}}{SS} \quad (1)$$

$$PD_k = \left(\frac{1}{M} \times \left(\sum_{i=1}^M \sqrt{(X_k - X_{Ei})^2 + (Y_k - Y_{Ei})^2} \right) + \frac{1}{N-1} \times \left(\sum_{j=1}^{N-1} \sqrt{(X_k - X_{Fj})^2 + (Y_k - Y_{Fj})^2} \right) \right) \times \frac{1}{SS}. \quad (2)$$

where, (X_k, Y_k) is the coordinate of k^{th} fog device and (X_{Ei}, Y_{Ei}) , (X_{Fj}, Y_{Fj}) are the coordinates of i^{th} EEG and j^{th} fog device respectively.

The processing delay of a fog device primarily depends on the clock speed of the fog processor. It is the total time taken to complete the assigned tasks and calculated as follows:

$$PrD_k = \frac{\mathcal{V}(\mathbb{T}_k)}{C_k^{avg}}. \quad (3)$$

where, $\mathcal{V}(\mathbb{T}_k)$ denotes the volume of EEG data assigned to k^{th} fog device and C_k^{avg} denotes the average clock speed.

The Average Queuing Delay (AQD_k) of k^{th} fog device is calculated as the average waiting time of a task in the buffer.

$$QD_k = \frac{BWT(T_k^1) + BWT(T_k^2) + \dots + BWT(T_k^n)}{n} \quad (4)$$

where n is total number of tasks, $BWT(T_k^i)$ is the buffer waiting time of i^{th} task and $T_k^1, T_k^2, \dots, T_k^n \in \mathbb{T}_k$

Definition 2: The Buffer Waiting Time ($BWT(T_k^n)$) of n^{th} task is the summation of processing time of $n - 1$ tasks.

$$BWT(T_k^n) = PD_k(T_k^1) + PD_k(T_k^2) + \dots + PD_k(T_k^{n-1}). \quad (5)$$

where $PD_k(T_k^i)$ is the processing time of i^{th} task assigned to k^{th} fog device and $BWT(T_k^1) = 0$

Finally, the Total Delay (TD_k) of k^{th} fog device is shown in Eq. 6.

$$TD_k = PD_k + PrD_k + AQD_k. \quad (6)$$

Further, D^{net} is the network delay which is calculated as the sum of total delay of all the fog devices.

$$D^{net} = \sum_{k=1}^N TD_k. \quad (7)$$

From Eq. 6 and Eq. 7, we have the fog delay index as

$$FDI_k = \frac{PD_k + PrD_k + AQD_k}{D^{Net}}. \quad (8)$$

Theorem 2: The Fog Performance Index (FPI_k) of k^{th} fog device is defined as the weighted summation of Idle Physical Memory (IPM_k) and Computational Efficiency (CE_k).

Proof: The performance of a fog device is generally dependent on its RAM and the processor components such as clock speed (C), Million Instructions Per Second (MIPS). The IPM_k of k^{th} fog device is calculated as:

$$IPM_k = \frac{RAM_k - \mathcal{V}(\mathbb{T}_k)}{RAM_k}. \quad (9)$$

The computational efficiency of k^{th} fog device is directly proportional to the MIPS and reciprocal of its clock speed. For the fog device having a minimum clock speed of C_k^{min} and a maximum clock speed of C_k^{max} , the average clock speed is calculated as

$$C_k^{avg} = \frac{C_k^{max} + C_k^{min}}{2}. \quad (10)$$

Consequently, CE_k is calculated as.

$$CE_k = \frac{MIPS_k \times 10^6}{C_k^{avg}}. \quad (11)$$

Let, γ be the relative weight of IPM_k and δ be the relative weight of CE_k . The FPI_k is summed from Eqs. 10 and 11 as

$$FPI_k = \gamma \times \frac{RAM_k - \mathcal{V}(\mathbb{T}_k)}{RAM_k} + \delta \times \frac{MIPS_k \times 10^6}{C_k^{avg}}. \quad (12)$$

where, $\gamma = \delta = 0.5$. This indicates the equal weightage of IPM_k and CE_k in calculating FPI_k .

2.2.2 Workflow

The workflow of the FH selection is given in Fig. 2. The aim is to select an optimal FH for the smooth operation of EEG tasks in the fog layer. Also, to ensure the uninterrupted operation in the fog layer, an additional fog device, called as Alternate Fog Head (AFH), is also chosen; which takes control of the FH in case of failure.

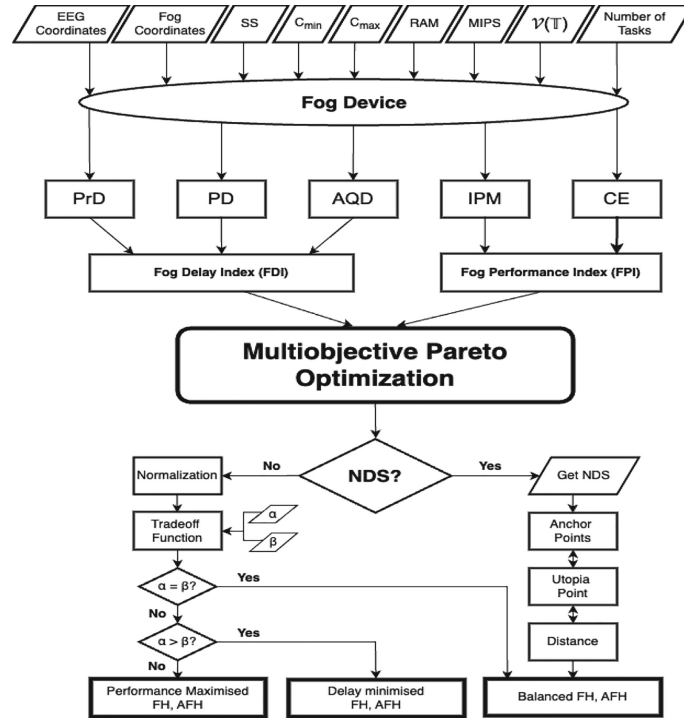


Fig. 2. Workflow diagram of the proposed 2DP-FHS technique.

The selection process of FH and AFH is described below:

Step 1: Each fog device is given inputs to calculate two values: FDI and FPI as given in Eq. 8 and Eq. 12.

Step 2: Multiobjective Pareto optimization is applied to separate Dominated Solution Space (DSS) and Non-dominated Solution Space (NSS). The solutions which have worse resultant values for either FDI or FPI compared to at least one solution in the solution space is called DSS. Similarly, NSS is the set of solutions that dominate at least one solution for both FDI and FPI. The NSS is generated based on the number of fog devices and their configurations.

Step 3: If NSS is present, then two Reference Points (RP) are selected as described. The first point is the least FDI and the second point is the most FPI among NSS. Next, a Utopia Point (UP) is calculated which is the intersection of two extreme values in NSS.

Step 4: If NSS is not present, then a tradeoff function, denoted by $f_{\alpha,\beta}(FDI, FPI)$, is calculated for each fog device which is the weighted sum of the normalized FDI and FPI values. Where α is the relative weight of FDI and β is the relative weight of FPI.

Step 5: Based on the number of non-dominated solutions in the NSS, there are three cases listed below.

Case I: If one non-dominated solution occurs, then that is selected as the FH, and AFH is selected as described in Step 4.

Case II: If NSS consists of more than one non-dominated solution, then FH and the AFH are selected based on the minimal distances from UP.

Case III: For the chosen α and β values, there are 3 sub-cases as listed below:

- If $\alpha > \beta$, solution impact: Delay Minimized
- If $\alpha < \beta$, solution impact: Performance Maximized
- If $\alpha = \beta$, solution impact: Balanced

3 Simulation and Result Analysis

The simulation work is carried out using Python v3.12 in MAC OS; having M1 Pro processor and 16 GB RAM; assuming the Bluetooth Low Energy as the medium of connection between the EEG and the fog devices in a maximum $200 \text{ m} \times 200 \text{ m}$ area. Moreover, a synthetic fog database is created with 8 distinct fog devices as given in the Table 1. Further, Table 2 describes the synthetic EEG data generated for processing and analysis.

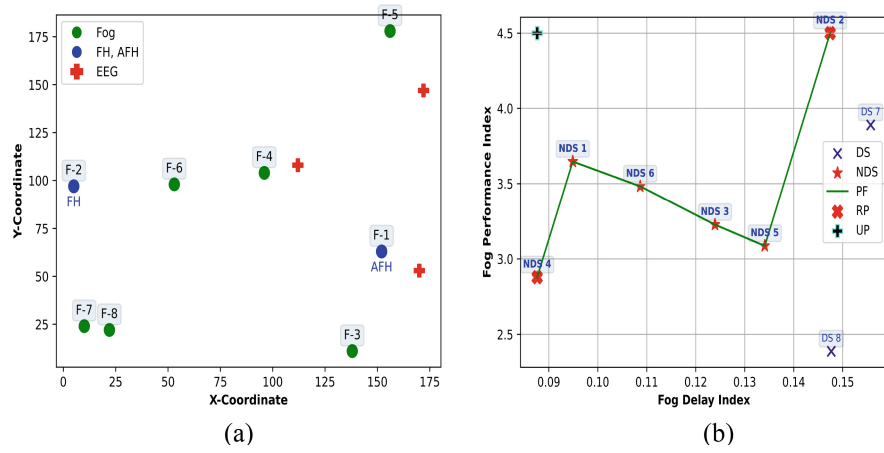
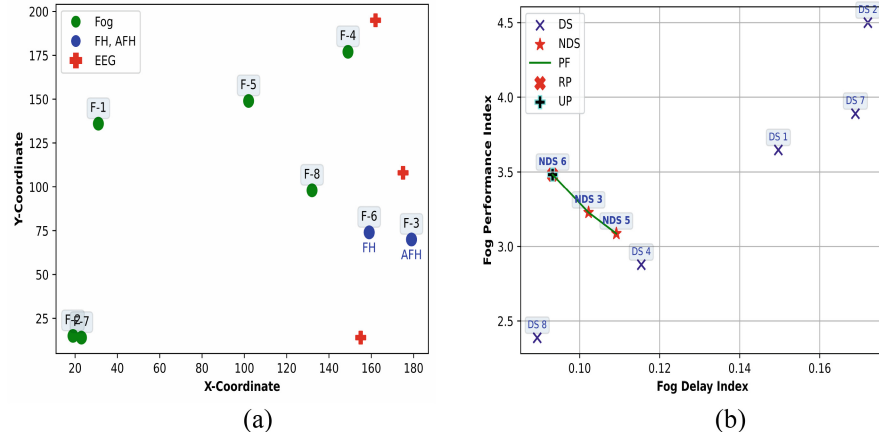
Table 1. Heterogeneous fog devices with their configurations.

Fog ID	C^{min} (in GHz)	C^{max} (in GHz)	RAM (in GB)	MIPS
1	1.43	1.43	4	9000
2	1.5	1.5	4	12000
3	1.4	1.9	8	9000
4	1.1	2.6	4	8800
5	1.8	2.84	8	12000
6	1.8	3.23	6	15000
7	2.5	3.4	4	20000
8	2.6	3.8	8	12080

Further, in this section, the authors have illustrated two different scenarios in selecting FH and AFH for NSS. Figure 3 shows the placement of the devices with the pareto solutions where UP is different from RP. However, Fig. 4 shows a different scenario where UP is same as RPs.

Table 2. Synthetic EEG dataset.

EEG No.	Channels	Sampling Frequency	Files	Data Volume
1	32	150 Hz	10	6.9945 Mb
2	16	170 Hz	4	5.24 Mb
3	64	250 Hz	6	15.42 Mb
4	32	230 Hz	5	7.95 Mb

**Fig. 3.** (a) Selection of FH (Fog 2) and AFH (Fog 1) for scenario-1, (b) UP is different from RPs**Fig. 4.** (a) Selection of FH (Fog 6) and AFH (Fog 3) for scenario-2, (b) UP and RPs are same

Furthermore, the selection of FH and AFH is shown in Table 4 corresponding to Table 3; where the fitness score is based on the tradeoff between FDI and FPI. The tradeoff function is calculated in Eq. 13 as follows.

$$f_{\alpha,\beta}(FDI, FPI) = \alpha * \frac{FDI_k}{\sum_{k=1}^N FDI_k} + \beta * \frac{FPI_k}{\sum_{k=1}^N FPI_k}. \quad (13)$$

Table 3. Synthetic fog dataset without non-dominated solution space.

Fog ID	FDI	FPI	Fog ID	FDI	FPI
1	0.4321	2.3232	5	0.6232	3.9272
2	0.5444	2.4444	6	0.1734	2.1111
3	0.8818	4.8734	7	0.1678	2.0676
4	0.5594	3.1212	8	0.8716	4.0256

Table 4. Selection of FH and AFH with variable weights

α	β	FH	AFH
1	0	7	6
0.9	0.1	7	6
0.8	0.2	6	7
0.7	0.3	1	2
0.6	0.4	2	4
0.5	0.5	2	4
0.4	0.6	4	2
0.3	0.7	5	8
0.2	0.8	8	5
0.1	0.9	3	8
0	1	3	8

In this work, the authors have emphasized the delay analysis with change in number of EEGs and fog devices. The result shown in Fig. 5 is generated from the average of 100 iterations. The graph represents the increase in average delay while increasing workloads and further decreasing with an increase in fog devices.

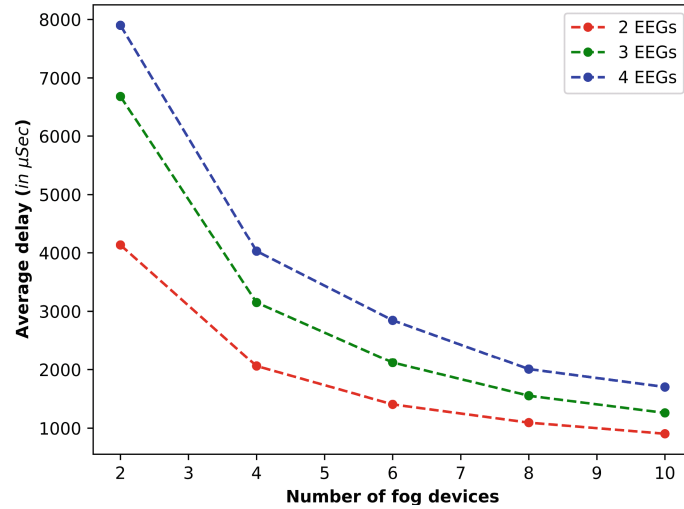


Fig. 5. The average delay analysis with increase in number of EEG devices and fog devices.

4 Conclusion

In this paper, the optimized fog head selection technique is discussed for EEG healthcare systems. Various factors such as distance, processor, number of devices, etc. are affecting in selection of optimal FH. The authors have considered all possible scenarios and unbiased fog characteristics to enhance the effectiveness of the 2DP-FHS technique. Additionally the results highlight the significance of this approach which works fine for heterogeneous fog devices and multiple EEG devices. In the future work, the authors have aimed to include more parameters considering energy and storage to select optimal fog devices.

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